

## AI-POWERED CAD ANALYSIS REPORT

Generated: 12 Aug 2026

Geometry to Process to Cost · Fully AI-reasoned

18

## PRCR002

277 × 223 × 182 mm · 1037.1 cm<sup>3</sup> · AI 2.800 kg / Steel 8.141 kg

Manufacturability: 18/100 · Confidence: High

## §1 — Geometry &amp; Part Summary

|                |                          |                  |                        |
|----------------|--------------------------|------------------|------------------------|
| Bounding Box   | 276.8 × 223.2 × 181.9 mm | Surface Area     | 1370.1 cm <sup>2</sup> |
| Volume         | 1037.11 cm <sup>3</sup>  | Weight (AI)      | 2.800 kg               |
| Weight (Steel) | 8.141 kg                 | Weight (Plastic) | 1.089 kg               |

## §2 — Detected Features

| Feature Type             | Count | Significance | Description                                                                                   |
|--------------------------|-------|--------------|-----------------------------------------------------------------------------------------------|
| Cylindrical bosses/holes | 132   | High         | 132 cylindrical faces, 21 estimated through-holes r<30mm                                      |
| Conical transitions      | 55    | Medium       | 55 conical faces suggest countersinks/tapered bosses — unusual for pure sheet metal           |
| Free-form surfaces       | 76    | High         | 76 B-spline faces indicating organic/blended geometry inconsistent with flat-pattern stamping |
| Torus features           | 34    | Medium       | 34 toroidal blends — typical of cast fillets, not bend radii                                  |
| Threaded features        | 1     | Medium       | Threaded bosses/holes detected requiring secondary tapping                                    |

## §3 — Material Analysis

AI-suggested

Primary Material: DC01 Mild Steel Sheet (55% confidence)

Stage-1 classifier forces sheet\_metal framing; deterministic strip/web math (3mm web) implies However, geometry (mean wall 34mm, max 169.95mm, 44 undercuts, 132 cylinders, bosses to 57mm radius) is far more consistent with a cast/machined steel part than a stamping. Following guidance to prefer steel over aluminium for a dense, thick-section, likely structural component absent an explicit aluminium signal.

Alternatives: High Strength Steel (25%) · Ductile Iron GJS500 (if actually cast) (15%)

## §4 — Process Recommendations

| Process                                | Commodity        | Confidence | Cycle Time (hr) | Reasoning                                                                                                                                                                                                                                                                                         |
|----------------------------------------|------------------|------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Progressive die stamping (steel)       | sheet_metal      | 45%        | 1.0680          | Per stage-1 pre-selection and mandated specialist framing; deterministic blank/strip/pitch values point to a 12-station progressive tool (1 blank + 25 bends + 1 pierce).                                                                                                                         |
| Sand or investment casting + machining | cast_and_machine | 40%        | 1.0680          | Geometry evidence (fill ratio 0.092, wall thickness range 0.45-169.95mm, 44 undercuts, 132 cylindrical/55 conical/34 toroidal faces, bosses to 57mm) is strongly inconsistent with a flat-pattern stamped part and instead matches a cast, thick-section component requiring secondary machining. |
| 5-axis CNC machining                   | machining        | 15%        | 1.0680          | High free-form content (76 B-spline faces) and undercuts could alternatively be produced by multi-axis milling if volumes were low; less likely at 200k units/yr.                                                                                                                                 |

## §5 — Manufacturability Risks (Score: 18/100)

| Severity | Feature / Area                           | Description                                                                                                                                                          | Recommended Action                                                                                                                       |
|----------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| High     | Undercuts (44 faces)                     | Large undercut count is incompatible with simple open/close stamping dies; would require complex cams/side-actions or is evidence the part is not truly sheet metal. | Re-verify process selection with actual drawing/photo; if stamped, redesign to eliminate undercuts or split into multiple formed shells. |
| High     | Wall thickness variation (0.45–169.95mm) | A 375:1 thickness ratio is impossible in a single-gauge sheet metal part.                                                                                            | Confirm with customer whether part is solid/cast; if sheet metal, uniform gauge must be specified.                                       |
| Medium   | 25 bend operations                       | High bend count in one progressive tool increases springback stack-up and die complexity.                                                                            | Simulate bend sequence and add relief features to control springback.                                                                    |
| Medium   | 21 pierced holes near bend lines         | Holes close to bend radii risk distortion during forming.                                                                                                            | Maintain hole-to-bend distance "e3 × thickness.                                                                                          |

## §6 — DFM Issues (sheet\_metal)

| Severity | Area                      | Description                                                                                                                                                                 | Impact                                                                                                                                          | Fix                                                                                                                                                 |
|----------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Critical | Process-geometry mismatch | Wall thickness spans 0.45mm to 169.95mm (mean 34mm) with 44 undercuts and 55 conical/34 toroidal faces — this is not achievable as a single-gauge stamped sheet metal part. | If forced into a progressive die, most of the part's volume/features cannot be formed, leading to tooling redesign, scrap, and project failure. | Confirm actual process with the customer's drawing/photo; if the part is genuinely solid/cast, switch commodityType to casting or cast_and_machine. |
| High     | Undercuts / side features | 44 undercut faces cannot be released from a simple 2-directional stamping die.                                                                                              | Requires cam-actuated slides or secondary forming stations, materially raising die cost beyond the £30k progressive-die estimate.               | Split geometry into multiple stamped/welded sub-assemblies or move to casting where side-cores are standard.                                        |
| High     | 25 bend stations          | High bend count within one strip increases cumulative springback and dimensional drift across stations.                                                                     | Increased first-off tuning time, higher scrap during ramp-up, tighter die maintenance intervals.                                                | Group bends into progressive stages with over-bend compensation and add coining stations at critical bends.                                         |
| Medium   | Hole proximity to         | 21 pierced holes, several near tight                                                                                                                                        | Hole tolerance drift and localized                                                                                                              | Relocate holes "e3 × thickness                                                                                                                      |

| Severity      | Area                     | Description                                                                     | Impact                                                                            | Fix                                                                               |
|---------------|--------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|               | <b>bends</b>             | bend radii, risk elongation/tearing during forming.                             | cracking, causing rework or scrap.                                                | from bend lines or pierce after forming in a secondary station.                   |
| <b>Medium</b> | <b>Threaded features</b> | Threads cannot be formed in-die and require secondary tapping or PEM insertion. | Adds secondary operation cost and cycle time not captured in base stamping cycle. | Add dedicated tapping/insert station to the line or use self-clinching fasteners. |

§7 — Cost Range & Suggested Inputs

| OPTIMISTIC   | MOST LIKELY  | CONSERVATIVE  |
|--------------|--------------|---------------|
| <b>£5.20</b> | <b>£7.80</b> | <b>£11.50</b> |

|                     |                                                                                                                                                       |            |                       |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------------------|
| Net Weight          | <b>8.141 kg</b>                                                                                                                                       | Material   | <b>mat-dc01</b>       |
| Recommended Process | <b>sheet_metal</b>                                                                                                                                    | Cycle Time | <b>1.0680 hr/part</b> |
| Setup Time          | <b>0.750 hr</b>                                                                                                                                       | Operations | <b>2 ops</b>          |
| Operations Detail   | <b>Progressive stamping (blank/pierce/bend x25)</b><br><b>(mach-haas-vf2, 0.0320 hr)</b><br><b>Secondary drilling/tapping (mach-drill, 0.1750 hr)</b> |            |                       |

Process-Specific Parameters

|                 |                      |
|-----------------|----------------------|
| Casting Subtype | <b>sand</b>          |
| Die/Mould Cost  | <b>£33,000</b>       |
| Die Life        | <b>5,000 shots</b>   |
| Cavities        | <b>1</b>             |
| Yield           | <b>65.0%</b>         |
| Flash Weight    | <b>0.800 kg</b>      |
| Yield           | <b>75.0%</b>         |
| Die Cost        | <b>£180,000</b>      |
| Strokes         | <b>8</b>             |
| Cavities        | <b>1</b>             |
| Wall Thickness  | <b>0 mm</b>          |
| Mould Cost      | <b>£200,000</b>      |
| Mould Life      | <b>500,000 shots</b> |
| Projected Area  | <b>0.0 cm²</b>       |

## §8 — AI Analysis Explanation

The part was routed to me as a sheet-metal specialist per Stage-1 classification, and I've populated all sheet-metal cost fields (thickness, die cost, die life, process, net weight, shear strength) using steel as the ground-truth material per the guidance to prefer steel over aluminium for dense, thick-section components absent an explicit aluminium signal. However, the underlying OCCT geometry (0.45-169.95mm wall range, fill ratio 0.092, 44 undercuts, 132 cylindrical + 55 conical + 34 toroidal faces, bosses to 57mm radius, threaded features) is strongly inconsistent with a true stamped sheet-metal part and instead matches a cast/machined steel component. I've flagged this mismatch as a Critical DFM issue and recommend the customer confirm the actual manufacturing route before tooling investment. All deterministic geometry, cycle-time, and parametric tooling figures were used verbatim as instructed.

## §9 — Analysis Limitations & Assumptions

1. Stage-1 sheet\_metal classification (conf 0.9) conflicts with strong geometric evidence for casting/machining (fill ratio 0.092, extreme wall thickness range, 44 undercuts) — recommend manual review of source drawing/photo.
2. Sheet gauge (1.5mm assumed) was inferred from the deterministic web-margin formula, not from a stated material spec — true gauge is unknown.
3. Net weight (8.141kg) is based on true solid volume at steel density, which is only meaningful if the part is genuinely a solid/cast steel body rather than a thin formed shell.
4. Die life and shear strength values are rule-of-thumb estimates for generic mild steel (DC01), not verified against an actual spec.

Stage-1 pre-selection: sheet\_metal (90%) · rotational\_moulding · casting